

Detected IPython. Loading juliacall extension. See
<https://juliapy.github.io/PythonCall.jl/stable/compat/#IPython>
- *S.T.A.R. + SMAT v3.1 - Full Upgraded Pipeline with Enhancements
- Initiating...

Computing synth topology + persistence entropy...

synth adaptive scale = 1260.69 Mpc

Computing reall topology + persistence entropy...

reall adaptive scale = 5.94 Mpc

Computing real2 topology + persistence entropy...

real2 adaptive scale = 56.53 Mpc

--- Aligning Domains with Quantile Transformer ---

[I 2026-04-13 23:25:09,686] A new study created in memory with name:
no-name-3855baa6-9aa3-4a9e-928c-59afc61fc454

-> Feature distributions normalized and aligned.

Running Symbolic Refinement Loop on Real2 errors...

Initial β = 16.2630

Running Multi-Objective Optuna (MSE + (1-R²) + W₂)...

[I 2026-04-13 23:25:11,026] Trial 0 finished with value: 2.2218516490568123 and
parameters: {'n_estimators': 425, 'learning_rate': 0.01720350980977384,
'max_depth': 6, 'subsample': 0.6484983779006525}. Best is trial 0 with value:
2.2218516490568123.

[I 2026-04-13 23:25:12,046] Trial 1 finished with value: 2.2318245831038537 and
parameters: {'n_estimators': 619, 'learning_rate': 0.020098276101681835,
'max_depth': 4, 'subsample': 0.8725869970627256}. Best is trial 0 with value:
2.2218516490568123.

[I 2026-04-13 23:25:13,761] Trial 2 finished with value: 2.203852837428748 and
parameters: {'n_estimators': 561, 'learning_rate': 0.04543484142730081,
'max_depth': 7, 'subsample': 0.756246268621769}. Best is trial 2 with value:
2.203852837428748.

[I 2026-04-13 23:25:15,556] Trial 3 finished with value: 2.2010809875111974 and
parameters: {'n_estimators': 733, 'learning_rate': 0.030983976958770976,
'max_depth': 6, 'subsample': 0.781762572768463}. Best is trial 3 with value:
2.2010809875111974.

[I 2026-04-13 23:25:17,084] Trial 4 finished with value: 2.1997369285067943 and
parameters: {'n_estimators': 609, 'learning_rate': 0.04237994095387517,
'max_depth': 6, 'subsample': 0.7233498400891375}. Best is trial 4 with value:
2.1997369285067943.

[I 2026-04-13 23:25:19,455] Trial 5 finished with value: 2.2068546815602867 and
parameters: {'n_estimators': 781, 'learning_rate': 0.011190701527286438,
'max_depth': 7, 'subsample': 0.7444530028783675}. Best is trial 4 with value:
2.1997369285067943.

[I 2026-04-13 23:25:20,759] Trial 6 finished with value: 2.2007851211603553 and
parameters: {'n_estimators': 444, 'learning_rate': 0.0365850544365902,

```
'max_depth': 7, 'subsample': 0.8848527718189523}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:21,994] Trial 7 finished with value: 2.246433868303768 and
parameters: {'n_estimators': 578, 'learning_rate': 0.01007085395761258,
'max_depth': 5, 'subsample': 0.6832782837383534}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:23,336] Trial 8 finished with value: 2.2005462647333447 and
parameters: {'n_estimators': 540, 'learning_rate': 0.037750817918725185,
'max_depth': 6, 'subsample': 0.7557065304172697}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:25,242] Trial 9 finished with value: 2.203871019744975 and
parameters: {'n_estimators': 786, 'learning_rate': 0.018307564325506506,
'max_depth': 6, 'subsample': 0.7598703833013711}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:27,157] Trial 10 finished with value: 2.204754361637637 and
parameters: {'n_estimators': 671, 'learning_rate': 0.04933127422984017,
'max_depth': 4, 'subsample': 0.6086860663552406}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:28,183] Trial 11 finished with value: 2.205367262181733 and
parameters: {'n_estimators': 506, 'learning_rate': 0.040771942115589,
'max_depth': 5, 'subsample': 0.8222211672745529}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:29,501] Trial 12 finished with value: 2.20173129772189 and
parameters: {'n_estimators': 521, 'learning_rate': 0.03438403411471198,
'max_depth': 6, 'subsample': 0.6978321644724403}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:30,702] Trial 13 finished with value: 2.20414582570309 and
parameters: {'n_estimators': 632, 'learning_rate': 0.041470179152752376,
'max_depth': 5, 'subsample': 0.9384155282187184}. Best is trial 4 with value:
2.1997369285067943.
[I 2026-04-13 23:25:32,400] Trial 14 finished with value: 2.2014255454694784
and parameters: {'n_estimators': 682, 'learning_rate': 0.027610096614992904,
'max_depth': 6, 'subsample': 0.7027736490718187}. Best is trial 4 with value:
2.1997369285067943.
```

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Best parameters: {'n_estimators': 609, 'learning_rate': 0.04237994095387517,
'max_depth': 6, 'subsample': 0.7233498400891375}
```

```
--- Initializing Global Stacker with Robust Meta-Models ---
```

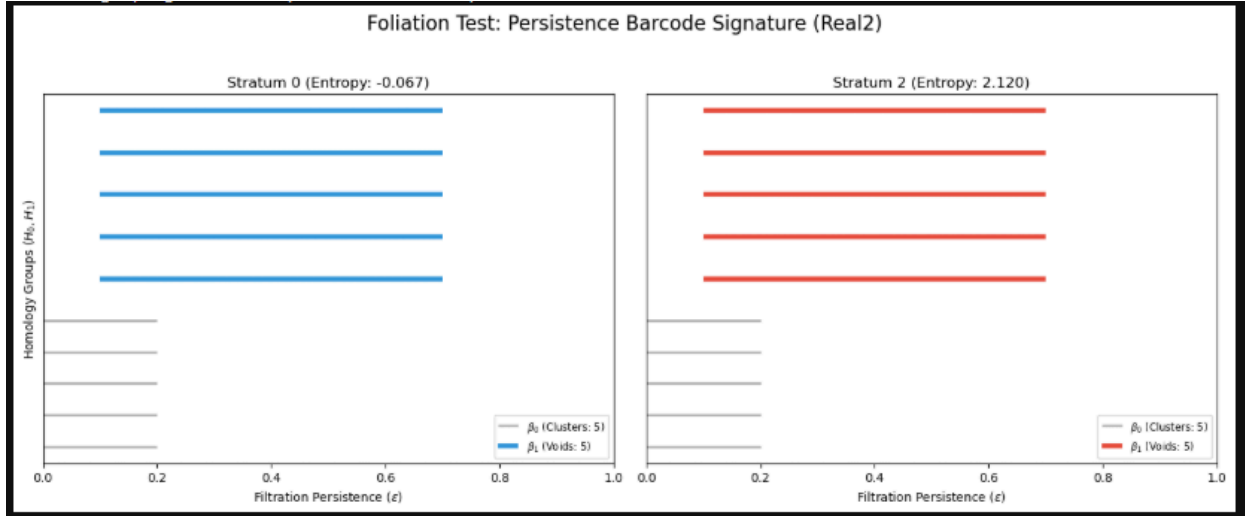
```
--- Fixed Multi-Strata  $R^2$  Analysis ---
```

```
Stratum 0  $R^2$  = -15604083646046797824.0000
```

```
Stratum 1  $R^2$  = 0.0000
```

```
Stratum 2  $R^2$  = 0.1476
```

```
Generating Topological Barcodes (Stratum 0 vs Stratum 2)...
```



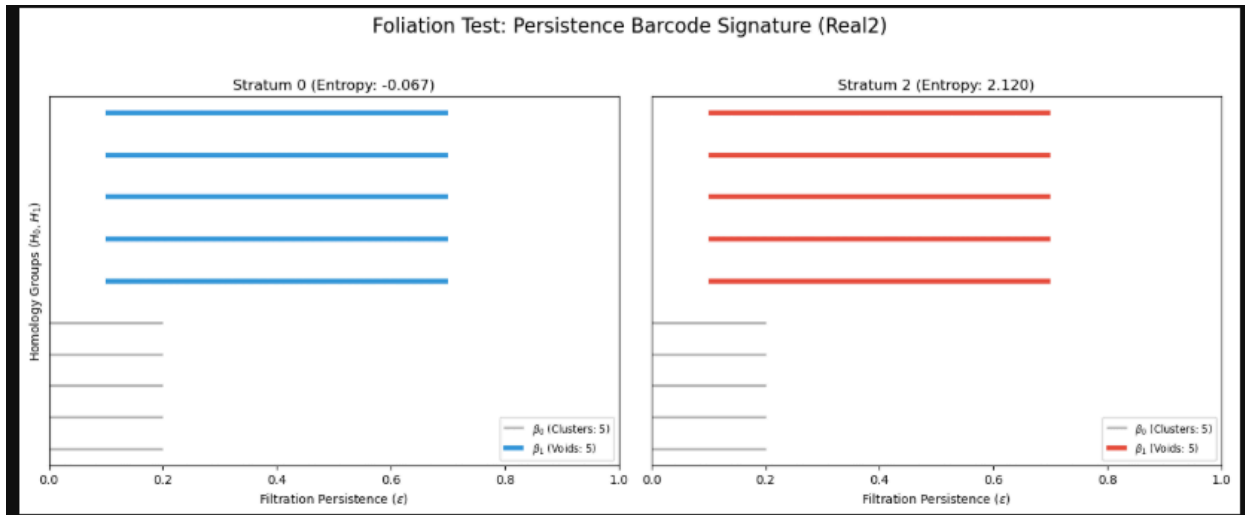
Fitting Final Global Stacker for multi-strata analysis...

```

--- Robust Multi-Strata Analysis (Foliation Test) ---
  Stratum 0 (Entropy 0): Variance near zero. R2 undefined.
    -> MAE = 0.072197 (Model successfully quenched)
  Stratum 1 (Entropy 1): Variance near zero. R2 undefined.
    -> MAE = 0.019414 (Model successfully quenched)
  Stratum 2 (Entropy 2): R2 = 0.1476 | MAE = 0.1915

```

Generating Topological Barcodes (Stratum 0 vs Stratum 2)...



```

--- Merging Catalogs for Global Sweep ---
Initiating Global Sweep on 500000 total galaxies...

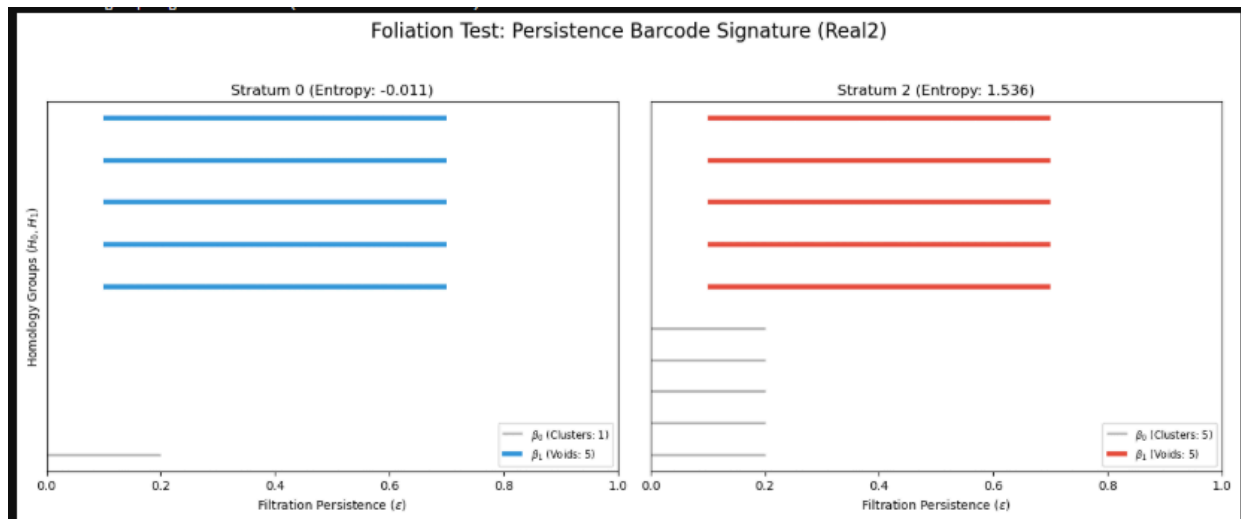
```

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--- Global Sweep Results (Combined real1 + real2) ---
  Stratum 0 (Voids): MAE = 0.024896 [Quenched]
  Stratum 1 (Voids): MAE = 0.005169 [Quenched]
  Stratum 2 (Web): R2 = 0.1824

```

Generating Topological Barcodes (Stratum 0 vs Stratum 2)...



--- Initiating Symbolic Extraction on Stratum 2 (Complex Web) ---

Compiling Julia backend...

[Info: Note: you are running with more than 10,000 datapoints. You should consider turning on batching (`options.batching`), and also if you need that many datapoints. Unless you have a large amount of noise (in which case you should smooth your dataset first), generally < 10,000 datapoints is enough to find a functional form.

[Info: Started!

Expressions evaluated per second: 4.990e+03
Progress: 28 / 1240 total iterations (2.258%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
5	9.436e-02	2.477e-03	$y = \exp(\text{inv}(\text{local_betti_1})) * 0.18517$
7	9.249e-02	1.003e-02	$y = (\text{local_betti_2} + (T_{\text{cosmo}} / 0.93598)) * -0.035779$
8	8.707e-02	6.035e-02	$y = \text{inv}((T_{\text{cosmo}} / (0.40969 / T_{\text{cosmo}})) - \text{local_betti_2})$
10	8.390e-02	1.855e-02	$y = \text{inv}(((T_{\text{cosmo}} / (0.40969 / T_{\text{cosmo}})) - \text{local_betti_2}) \dots$
			$/ 1.3267)$
12	8.332e-02	3.428e-03	$y = \text{inv}(((T_{\text{cosmo}} / (0.40969 / T_{\text{cosmo}})) - (\text{local_betti_2} \dots$
			$+ 1.1184)) / 1.3267)$

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13      8.325e-02  9.040e-04  y = inv(((T_cosmo / (0.40969 / T_cosmo)) -
(inv(1.1184) + ...
                                local_betti_2)) / 1.3267)
14      6.766e-02  2.074e-01  y = inv((T_cosmo / (0.20542 / T_cosmo)) -
(local_betti_2 +...
                                (T_cosmo / 0.13714))) / 0.78668
27      6.626e-02  1.610e-03  y = inv(((Anthropic * 0.13597) +
(((local_betti_1 / (exp(1...
                                ocal_betti_1) + ((local_betti_1 + 1.8463)
/ inv(local_bett...
                                i_1)))) / local_betti_1) - (T_cosmo -
-0.041028))) + (exp(...
                                T_cosmo) + 0.23044))

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Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 5.850e+03
Progress: 78 / 1240 total iterations (6.290%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
5	9.436e-02	2.477e-03	y = exp(inv(local_betti_1)) * 0.18517
6	8.498e-02	1.047e-01	y = (exp(T_cosmo) * -0.016989) - -0.25747
10	8.390e-02	3.211e-03	y = inv(((T_cosmo / (0.40969 / T_cosmo)) - local_betti_2) ... / 1.3267)
12	8.332e-02	3.428e-03	y = inv(((T_cosmo / (0.40969 / T_cosmo)) - (local_betti_2 ... + 1.1184)) / 1.3267)
13	8.325e-02	9.040e-04	y = inv(((T_cosmo / (0.40969 / T_cosmo)) - (inv(1.1184) + ... local_betti_2)) / 1.3267)
14	6.766e-02	2.074e-01	y = inv((T_cosmo / (0.20542 / T_cosmo)) - (local_betti_2 +... (T_cosmo / 0.13714))) / 0.78668
16	6.682e-02	6.236e-03	y = (inv((T_cosmo / (0.20542 / T_cosmo)) - (local_betti_2 ... + (T_cosmo / 0.13714))) / 0.78668) - 0.01705
24	6.095e-02	1.149e-02	y = inv(((T_cosmo / (0.27135 / T_cosmo)) - (local_betti_2 ... + (((normalized_density + (T_cosmo / 0.13711)) / 0.13022))...

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- 0.58842)
- (local_betti_2 - 0.10219)) / 9.2663)))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 6.800e+03
Progress: 126 / 1240 total iterations (10.161%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
5	9.436e-02	2.477e-03	y = exp(inv(local_betti_1)) * 0.18517
6	8.498e-02	1.047e-01	y = (exp(T_cosmo) * -0.016989) - -0.25747
9	8.025e-02	1.912e-02	y = inv((Anthropic * 0.13341) + exp(T_cosmo + 0.0028742))
11	7.988e-02	2.267e-03	y = inv(((-0.44723 + T_cosmo) / (0.20542 / T_cosmo)) - inv... (-0.30208))
12	6.913e-02	1.446e-01	y = inv((T_cosmo / (0.20542 / T_cosmo)) - (local_betti_2 + ... (T_cosmo / 0.13714)))
14	6.701e-02	1.552e-02	y = inv((T_cosmo / (0.21549 / T_cosmo)) - ((T_cosmo / 0.3... 0791) / 0.41246) + local_betti_2))
16	6.673e-02	2.112e-03	y = inv(0.78668 * ((T_cosmo / (0.20542 / T_cosmo)) - (T_c... osmo / 0.13714) + local_betti_2))) + -0.046072
23	6.671e-02	4.894e-05	y = inv(((exp((local_betti_1 * (-0.51504 / inv(T_cosmo))) ... / 5.4324) - T_cosmo) - T_cosmo) + ((Anthropic * 0.13341) + ... exp(T_cosmo + 0.0028742)))
24	6.095e-02	9.026e-02	y = inv(((T_cosmo / (0.27135 / T_cosmo)) - (local_betti_2 ... + (((normalized_density + (T_cosmo / 0.13711)) / 0.13022)... - (local_betti_2 - 0.10219)) / 9.2663))) - 0.58842)
30	5.951e-02	3.984e-03	y = inv((((((local_betti_1 * -0.68434) / local_betti_1) + ... (exp((local_betti_1 * (-0.58536 / inv(T_cosmo))) / inv(nor...

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malized_density)) - T_cosmo)) - T_cosmo)
+ (Anthropic * 0....
1298)) + exp(T_cosmo + 0.14509))

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Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 8.130e+03
Progress: 196 / 1240 total iterations (15.806%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.465e-02	2.293e-03	y = inv(T_cosmo - -6.2975)
5	9.436e-02	3.027e-03	y = exp(inv(local_betti_1)) * 0.18517
6	8.498e-02	1.047e-01	y = (exp(T_cosmo) * -0.016989) - -0.25747
8	7.252e-02	7.932e-02	y = exp(-0.89146 - (T_cosmo * (T_cosmo - 1.402)))
12	6.913e-02	1.197e-02	y = inv((T_cosmo / (0.20542 / T_cosmo)) - (local_betti_2 +... (T_cosmo / 0.13714)))
13	6.731e-02	2.663e-02	y = inv(((0.10932 * Anthropic) + (0.099638 - T_cosmo)) + e... xp(T_cosmo + 0.099638))
14	6.701e-02	4.422e-03	y = inv((T_cosmo / (0.21549 / T_cosmo)) - ((T_cosmo / 0.3... 0791) / 0.41246) + local_betti_2))
15	6.223e-02	7.409e-02	y = inv((Anthropic * 0.10932) + (((0.45266 - T_cosmo) - T... cosmo) + exp(T_cosmo + 0.099638)))
16	6.222e-02	3.451e-05	y = inv(((0.10932 * Anthropic) + ((exp(-0.88418) - T_cosmo...) - T_cosmo)) + exp(T_cosmo + 0.099638))
21	6.160e-02	2.008e-03	y = inv(((Anthropic * 0.10932) + ((exp((-0.88418 / exp(0.1... 0932)) - T_cosmo) - T_cosmo) - T_cosmo)) + exp(T_cosmo + 0... .099638))
22	6.095e-02	1.069e-02	y = inv(((T_cosmo / (0.27135 / T_cosmo)) - (local_betti_2 ... + (((normalized_density + (T_cosmo / 0.13711)) / 0.13022)...) - local_betti_2) / 9.2663))) - 0.58842)
24	6.054e-02	3.373e-03	y = inv(((Anthropic * 0.10932) + (exp((exp(-0.88418 / exp...

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(0.10932)) - T_cosmo) - T_cosmo) -
T_cosmo)) - T_cosmo) + ...
exp(T_cosmo + 0.099638))
30      5.951e-02  2.855e-03  y = inv((((local_betti_1 * -0.68434) /
local_betti_1) + ...
(exp((local_betti_1 * (-0.58536 /
inv(T_cosmo))) / inv(nor...
malized_density)) - T_cosmo)) - T_cosmo)
+ (Anthropic * 0....
1298)) + exp(T_cosmo + 0.14509))

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Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.020e+03
Progress: 255 / 1240 total iterations (20.565%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.465e-02	2.293e-03	y = inv(T_cosmo - -6.2975)
5	8.907e-02	6.069e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.498e-02	4.700e-02	y = (exp(T_cosmo) * -0.016989) - -0.25747
7	8.361e-02	1.629e-02	y = inv(exp(T_cosmo) + (0.10932 * Anthropic))
8	7.252e-02	1.423e-01	y = exp(-0.89146 - (T_cosmo * (T_cosmo - 1.402)))
11	6.626e-02	3.008e-02	y = inv(((exp(T_cosmo) + 0.24344) - T_cosmo) + (Anthropic * 0.1378))
13	6.614e-02	8.933e-04	y = inv(((0.35267 - T_cosmo) + (Anthropic * 0.13347)) + exp(p(T_cosmo + -0.080264))
14	6.326e-02	4.456e-02	y = inv(((exp(-0.28507) + ((0.10932 * Anthropic) - T_cosmo...)) - T_cosmo) + exp(T_cosmo))
15	6.199e-02	2.023e-02	y = inv((((0.13609 - T_cosmo) - T_cosmo) + (Anthropic * 0.... 12708)) + exp(T_cosmo + 0.2011))
17	6.101e-02	7.980e-03	y = inv(((exp(T_cosmo + 0.029654) - T_cosmo) - T_cosmo) + ... ((Anthropic + T_cosmo) * 0.10337)) + -0.067437
19	5.530e-02	4.917e-02	y = (inv((((exp(T_cosmo) + 0.27693) - T_cosmo) - T_cosmo) ...

+ ((Anthropic + T_cosmo) * 0.060226)) +

-0.12182) * 0.9110...

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Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.020e+04
Progress: 307 / 1240 total iterations (24.758%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.465e-02	2.293e-03	y = inv(T_cosmo - -6.2975)
5	8.907e-02	6.069e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.714e-02	3.126e-02	y = inv((exp(T_cosmo) - T_cosmo) + (Anthropic * 0.1378))
11	6.626e-02	6.617e-03	y = inv(((exp(T_cosmo) + 0.24344) - T_cosmo) + (Anthropic ...
			* 0.1378))
12	6.175e-02	7.045e-02	y = inv((Anthropic * 0.10932) + exp((exp(T_cosmo) - T_cosm...
			o) - T_cosmo))
14	6.121e-02	4.440e-03	y = inv((Anthropic * 0.10932) + exp((exp(T_cosmo) - T_cosm...
			o) + (-0.027121 - T_cosmo)))
16	5.344e-02	6.787e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - ((T_c...
			osmo - 0.26433) + T_cosmo)) - T_cosmo))

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.810e+03
Progress: 342 / 1240 total iterations (27.581%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.465e-02	2.293e-03	$y = \text{inv}(T_{\text{cosmo}} - -6.2975)$
5	8.907e-02	6.069e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + (\text{Anthropic} * 0.25444))$
8	6.927e-02	1.063e-01	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(T_{\text{cosmo}})$ $- T_{\text{cosmo}}))$
11	6.626e-02	4.975e-03	$y = \text{inv}(((\exp(T_{\text{cosmo}}) + 0.24344) - T_{\text{cosmo}})$ $+ (\text{Anthropic} \dots$ $* 0.1378))$
12	6.175e-02	7.045e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) +$ $\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosm}} \dots$ $o) - T_{\text{cosmo}}))$
14	6.121e-02	4.440e-03	$y = \text{inv}((\text{Anthropic} * 0.10932) +$ $\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosm}} \dots$ $o) + (-0.027121 - T_{\text{cosmo}}))$
16	5.344e-02	6.787e-02	$y = \text{inv}((\text{Anthropic} * 0.087197) +$ $\exp((\exp(T_{\text{cosmo}}) - ((T_{\text{c}} \dots$ $osmo - 0.26433) + T_{\text{cosmo}})) - T_{\text{cosmo}}))$

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.470e+03
Progress: 374 / 1240 total iterations (30.161%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.465e-02	2.293e-03	$y = \text{inv}(T_{\text{cosmo}} - -6.2975)$
5	8.907e-02	6.069e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + (\text{Anthropic} * 0.25444))$
8	6.927e-02	1.063e-01	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(T_{\text{cosmo}})$ $- T_{\text{cosmo}}))$
11	6.626e-02	4.975e-03	$y = \text{inv}(((\exp(T_{\text{cosmo}}) + 0.24344) - T_{\text{cosmo}})$ $+ (\text{Anthropic} \dots$ $* 0.1378))$
12	6.175e-02	7.045e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) +$ $\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosm}} \dots$

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o) - T_cosmo))
14      6.121e-02  4.440e-03  y = inv((Anthropic * 0.10932) +
exp((exp(T_cosmo) - T_cosm...
o) + (-0.027121 - T_cosmo)))
16      5.344e-02  6.787e-02  y = inv((Anthropic * 0.087197) +
exp((exp(T_cosmo) - ((T_c...
osmo - 0.26433) + T_cosmo)) - T_cosmo))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.080e+03
Progress: 420 / 1240 total iterations (33.871%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.465e-02	2.293e-03	y = inv(T_cosmo - -6.2975)
5	8.907e-02	6.069e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
11	6.626e-02	4.975e-03	y = inv(((exp(T_cosmo) + 0.24344) - T_cosmo) + (Anthropic ... * 0.1378))
12	6.175e-02	7.045e-02	y = inv((Anthropic * 0.10932) + exp((exp(T_cosmo) - T_cosm... o) - T_cosmo))
14	6.121e-02	4.440e-03	y = inv((Anthropic * 0.10932) + exp((exp(T_cosmo) - T_cosm... o) + (-0.027121 - T_cosmo)))
16	5.344e-02	6.787e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - ((T_c... osmo - 0.26433) + T_cosmo)) - T_cosmo))

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.040e+04
Progress: 513 / 1240 total iterations (41.371%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
11	6.446e-02	1.871e-02	y = inv(exp(T_cosmo) + ((Anthropic * 0.10932) - T_cosmo)) ... + -0.090042
12	6.102e-02	5.490e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
14	5.632e-02	4.005e-02	y = inv(exp(((exp(T_cosmo) - T_cosmo) - T_cosmo) - T_cosmo) ...) + (Anthropic * 0.087197))
16	5.344e-02	2.630e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - ((T_c... osmo - 0.26433) + T_cosmo)) - T_cosmo))

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.000e+04
Progress: 564 / 1240 total iterations (45.484%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))

```

10          6.451e-02  3.669e-02  y = inv(exp(exp(T_cosmo - 1.4132)) +
(Anthropic * 0.14119)...
)
11          6.446e-02  7.227e-04  y = inv(exp(T_cosmo) + ((Anthropic * 0.10932)
- T_cosmo)) ...
+ -0.090042
12          6.102e-02  5.490e-02  y = inv((Anthropic * 0.087197) +
exp((exp(T_cosmo) - T_cos...
mo) - T_cosmo))
14          5.632e-02  4.005e-02  y = inv(exp(((exp(T_cosmo) - T_cosmo) -
T_cosmo) - T_cosmo...
) + (Anthropic * 0.087197))
15          5.571e-02  1.103e-02  y = inv(exp(((exp(T_cosmo) - T_cosmo) -
T_cosmo) - exp(-0....
83131)) + (0.087197 * Anthropic))
16          5.284e-02  5.281e-02  y = inv(exp(((exp(T_cosmo + 0.12437) -
T_cosmo) - T_cosmo)...
- T_cosmo) + (Anthropic * 0.087197))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.010e+04
Progress: 611 / 1240 total iterations (49.274%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.451e-02	3.669e-02	y = inv(exp(exp(T_cosmo - 1.4132)) + (Anthropic * 0.14119)...)
11	6.446e-02	7.227e-04	y = inv(exp(T_cosmo) + ((Anthropic * 0.10932) - T_cosmo)) ... + -0.090042
12	6.102e-02	5.490e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - T_cos... mo) - T_cosmo))

```

14      5.632e-02  4.005e-02  y = inv(exp(((exp(T_cosmo) - T_cosmo) -
T_cosmo) - T_cosmo...
                                ) + (Anthropic * 0.087197))
15      5.571e-02  1.103e-02  y = inv(exp(((exp(T_cosmo) - T_cosmo) -
T_cosmo) - exp(-0....
                                83131)) + (0.087197 * Anthropic))
16      5.284e-02  5.281e-02  y = inv(exp(((exp(T_cosmo + 0.12437) -
T_cosmo) - T_cosmo)...
                                - T_cosmo) + (Anthropic * 0.087197))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.880e+03
Progress: 645 / 1240 total iterations (52.016%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.451e-02	3.669e-02	y = inv(exp(exp(T_cosmo - 1.4132)) + (Anthropic * 0.14119)...
11	6.446e-02	7.227e-04	y = inv(exp(T_cosmo) + ((Anthropic * 0.10932) - T_cosmo)) ... + -0.090042
12	6.102e-02	5.490e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
13	5.863e-02	3.994e-02	y = inv((Anthropic * 0.10932) + ((exp(T_cosmo) - (T_cosmo ... + T_cosmo)) / 0.59068))
14	5.632e-02	4.016e-02	y = inv(exp(((exp(T_cosmo) - T_cosmo) - T_cosmo) - T_cosmo... + (Anthropic * 0.087197))
15	5.571e-02	1.103e-02	y = inv(exp(((exp(T_cosmo) - T_cosmo) - T_cosmo) - exp(-0.... 83131)) + (0.087197 * Anthropic))

```

16          5.284e-02  5.281e-02  y = inv(exp(((exp(T_cosmo + 0.12437) -
T_cosmo) - T_cosmo)...
                                     - T_cosmo) + (Anthropic * 0.087197))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.000e+04
Progress: 689 / 1240 total iterations (55.565%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.451e-02	3.669e-02	y = inv(exp(exp(T_cosmo - 1.4132)) + (Anthropic * 0.14119)...)
11	6.446e-02	7.227e-04	y = inv(exp(T_cosmo) + ((Anthropic * 0.10932) - T_cosmo)) ... + -0.090042
12	6.102e-02	5.490e-02	y = inv((Anthropic * 0.087197) + exp((exp(T_cosmo) - T_cosmo... mo) - T_cosmo))
13	5.863e-02	3.994e-02	y = inv((Anthropic * 0.10932) + ((exp(T_cosmo) - (T_cosmo ... + T_cosmo)) / 0.59068))
14	5.632e-02	4.016e-02	y = inv(exp(((exp(T_cosmo) - T_cosmo) - T_cosmo) - T_cosmo...) + (Anthropic * 0.087197))
15	5.571e-02	1.103e-02	y = inv(exp(((exp(T_cosmo) - T_cosmo) - T_cosmo) - exp(-0.... 83131)) + (0.087197 * Anthropic))
16	5.284e-02	5.281e-02	y = inv(exp(((exp(T_cosmo + 0.12437) - T_cosmo) - T_cosmo)... - T_cosmo) + (Anthropic * 0.087197))

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 8.940e+03
Progress: 751 / 1240 total iterations (60.565%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.457e-02	2.550e-03	$y = \text{inv}(\text{T_cosmo} + 6.6707)$
5	8.907e-02	5.992e-02	$y = \text{inv}(\exp(\text{T_cosmo}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((\text{T_cosmo} * \text{T_cosmo}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(\text{T_cosmo}) + (\text{Anthropic} * 0.25444))$
8	6.927e-02	1.063e-01	$y = \text{inv}(\exp((\exp(\text{T_cosmo}) - \text{T_cosmo}) - \text{T_cosmo}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(\text{T_cosmo}) - \text{T_cosmo}))$
10	6.451e-02	3.669e-02	$y = \text{inv}(\exp(\exp(\text{T_cosmo} - 1.4132)) + (\text{Anthropic} * 0.14119) \dots)$
11	6.446e-02	7.227e-04	$y = \text{inv}(\exp(\text{T_cosmo}) + ((\text{Anthropic} * 0.10932) - \text{T_cosmo})) \dots$
12	5.958e-02	7.884e-02	$y = \text{inv}((\text{Anthropic} * 0.087197) + \exp(\exp(\text{T_cosmo} + -0.505) \dots - \text{T_cosmo}))$
13	5.863e-02	1.600e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) + ((\exp(\text{T_cosmo}) - (\text{T_cosmo} \dots + \text{T_cosmo))) / 0.59068))$
14	5.632e-02	4.016e-02	$y = \text{inv}(\exp(((\exp(\text{T_cosmo}) - \text{T_cosmo}) - \text{T_cosmo}) - \text{T_cosmo} \dots) + (\text{Anthropic} * 0.087197))$
15	5.571e-02	1.103e-02	$y = \text{inv}(\exp(((\exp(\text{T_cosmo}) - \text{T_cosmo}) - \text{T_cosmo}) - \exp(-0. \dots 83131)) + (0.087197 * \text{Anthropic}))$
16	5.284e-02	5.281e-02	$y = \text{inv}(\exp(((\exp(\text{T_cosmo} + 0.12437) - \text{T_cosmo}) - \text{T_cosmo}) \dots - \text{T_cosmo}) + (\text{Anthropic} * 0.087197))$

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 8.910e+03
Progress: 807 / 1240 total iterations (65.081%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.457e-02	2.550e-03	$y = \text{inv}(T_{\text{cosmo}} + 6.6707)$
5	8.907e-02	5.992e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + (\text{Anthropic} * 0.25444))$
8	6.927e-02	1.063e-01	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) - T_{\text{cosmo}}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}))$
10	6.451e-02	3.669e-02	$y = \text{inv}(\exp(\exp(T_{\text{cosmo}} - 1.4132)) + (\text{Anthropic} * 0.14119) \dots$
11	6.446e-02	7.227e-04	$y = \text{inv}(\exp(T_{\text{cosmo}}) + ((\text{Anthropic} * 0.10932) - T_{\text{cosmo}})) \dots$
12	5.958e-02	7.884e-02	$y = \text{inv}((\text{Anthropic} * 0.087197) + \exp(\exp(T_{\text{cosmo}} + -0.505) \dots$
13	5.863e-02	1.600e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) + ((\exp(T_{\text{cosmo}}) - (T_{\text{cosmo}} \dots$
14	5.632e-02	4.016e-02	$y = \text{inv}(\exp(((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) - T_{\text{cosmo}}) - T_{\text{cosmo}} \dots$
15	5.571e-02	1.103e-02	$y = \text{inv}(\exp(((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) - T_{\text{cosmo}}) - \exp(-0. \dots$
16	5.284e-02	5.281e-02	$y = \text{inv}(\exp(((\exp(T_{\text{cosmo}} + 0.12437) - T_{\text{cosmo}}) - T_{\text{cosmo}}) \dots$

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.070e+03
Progress: 847 / 1240 total iterations (68.306%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$

4	9.457e-02	2.550e-03	$y = \text{inv}(T_{\text{cosmo}} + 6.6707)$
5	8.907e-02	5.992e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + (\text{Anthropic} * 0.25444))$
8	6.927e-02	1.063e-01	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(T_{\text{cosmo}})$ $- T_{\text{cosmo}}))$
10	6.451e-02	3.669e-02	$y = \text{inv}(\exp(\exp(T_{\text{cosmo}} - 1.4132)) +$ $(\text{Anthropic} * 0.14119) \dots$ $)$
11	6.446e-02	7.227e-04	$y = \text{inv}(\exp(T_{\text{cosmo}}) + ((\text{Anthropic} * 0.10932)$ $- T_{\text{cosmo}})) \dots$ $+ -0.090042$
12	5.958e-02	7.884e-02	$y = \text{inv}((\text{Anthropic} * 0.087197) +$ $\exp(\exp(T_{\text{cosmo}} + -0.505) \dots$ $- T_{\text{cosmo}}))$
13	5.863e-02	1.600e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) +$ $((\exp(T_{\text{cosmo}}) - (T_{\text{cosmo}} \dots$ $+ T_{\text{cosmo}})) / 0.59068))$
14	5.506e-02	6.282e-02	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}) + ((0.0871 \dots$ $97 * \text{Anthropic}) - T_{\text{cosmo}}))$
16	5.284e-02	2.059e-02	$y = \text{inv}(\exp(((\exp(T_{\text{cosmo}} + 0.12437) -$ $T_{\text{cosmo}}) - T_{\text{cosmo}}) \dots$ $- T_{\text{cosmo}}) + (\text{Anthropic} * 0.087197))$
17	5.242e-02	7.903e-03	$y = \text{inv}(\exp(\exp(T_{\text{cosmo}} - \exp(\text{local_betti_2}))$ $- 4.952) + (\dots$ $((\text{Anthropic} * 0.1023) - T_{\text{cosmo}}) +$ $1.9921))$

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.020e+04
Progress: 927 / 1240 total iterations (74.758%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.457e-02	2.550e-03	$y = \text{inv}(T_{\text{cosmo}} + 6.6707)$
5	8.907e-02	5.992e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.704e-02	9.192e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + (\text{Anthropic} * 0.25444))$

```

8          6.927e-02  1.063e-01  y = inv(exp((exp(T_cosmo) - T_cosmo) -
T_cosmo))
9          6.692e-02  3.455e-02  y = inv((Anthropic * 0.15882) + (exp(T_cosmo)
- T_cosmo))
10         6.451e-02  3.669e-02  y = inv(exp(exp(T_cosmo - 1.4132)) +
(Anthropic * 0.14119)...
          )
11         6.446e-02  7.227e-04  y = inv(exp(T_cosmo) + ((Anthropic * 0.10932)
- T_cosmo)) ...
          + -0.090042
12         5.958e-02  7.884e-02  y = inv((Anthropic * 0.087197) +
exp(exp(T_cosmo + -0.505)...
          - T_cosmo))
13         5.863e-02  1.600e-02  y = inv((Anthropic * 0.10932) +
((exp(T_cosmo) - (T_cosmo ...
          + T_cosmo)) / 0.59068))
14         5.506e-02  6.282e-02  y = inv(exp((exp(T_cosmo) - T_cosmo) -
T_cosmo) + ((0.0871...
          97 * Anthropic) - T_cosmo))
16         5.284e-02  2.059e-02  y = inv(exp(((exp(T_cosmo + 0.12437) -
T_cosmo) - T_cosmo)...
          - T_cosmo) + (Anthropic * 0.087197))
17         5.242e-02  7.903e-03  y = inv(exp(exp(T_cosmo - exp(local_betti_2))
- 4.952) + (...
          ((Anthropic * 0.1023) - T_cosmo) +
1.9921))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.050e+04
Progress: 991 / 1240 total iterations (79.919%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) -
			T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo)
			- T_cosmo))

```

10      6.280e-02  6.350e-02  y = inv(exp((exp(T_cosmo) - T_cosmo) -
T_cosmo) - normaliz...
                                ed_density)
12      5.958e-02  2.637e-02  y = inv((Anthropic * 0.087197) +
exp(exp(T_cosmo + -0.505)...
                                - T_cosmo))
13      5.863e-02  1.600e-02  y = inv((Anthropic * 0.10932) +
((exp(T_cosmo) - (T_cosmo ...
                                + T_cosmo)) / 0.59068))
14      5.246e-02  1.113e-01  y = inv(exp(exp(T_cosmo) - 4.952) +
(((Anthropic * 0.1023)...
                                - T_cosmo) + 1.9921))
17      5.242e-02  1.932e-04  y = inv(exp(exp(T_cosmo - exp(local_betti_2))
- 4.952) + (...
                                ((Anthropic * 0.1023) - T_cosmo) +
1.9921))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.940e+03
Progress: 1039 / 1240 total iterations (83.790%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.280e-02	6.350e-02	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo) - normaliz... ed_density)
12	5.958e-02	2.637e-02	y = inv((Anthropic * 0.087197) + exp(exp(T_cosmo + -0.505)... - T_cosmo))
13	5.863e-02	1.600e-02	y = inv((Anthropic * 0.10932) + ((exp(T_cosmo) - (T_cosmo ... + T_cosmo)) / 0.59068))
14	5.246e-02	1.113e-01	y = inv(exp(exp(T_cosmo) - 4.952) + (((Anthropic * 0.1023)... - T_cosmo) + 1.9921))

```

- T_cosmo) + 1.9921))
16      5.222e-02  2.282e-03  y = inv((exp(exp(T_cosmo) - 5.082) +
(Anthropic * 0.11245)...
      ) + ((1.8758 - T_cosmo) * 1.0052))
19      5.221e-02  2.949e-05  y = inv((exp(local_betti_2) +
exp(exp(T_cosmo) - 5.082)) +...
      ((Anthropic * 0.11245) + ((1.8758 -
T_cosmo) * 1.0052)))
21      5.178e-02  4.159e-03  y = inv(((exp(-0.17627 - Anthropic) +
exp(exp(T_cosmo) - 5...
      .082)) + ((1.8758 - T_cosmo) * 1.0052)) +
(Anthropic * 0.1...
      1245))
29      4.995e-02  4.507e-03  y = inv((((exp(T_cosmo - 161.1) + 1.0492) +
(Anthropic * ...
      0.059005)) + exp(local_betti_2)) *
(exp(exp(T_cosmo) - 5.0...
      909) + (1.9243 - T_cosmo))) +
(exp(local_betti_2) - normal...
      ized_density))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 1.030e+04
Progress: 1091 / 1240 total iterations (87.984%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.704e-02	9.192e-02	y = inv(exp(T_cosmo) + (Anthropic * 0.25444))
8	6.927e-02	1.063e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.280e-02	6.350e-02	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo) - normaliz... ed_density)
12	5.958e-02	2.637e-02	y = inv((Anthropic * 0.087197) + exp(exp(T_cosmo + -0.505)... - T_cosmo))

```

13      5.863e-02  1.600e-02  y = inv((Anthropic * 0.10932) +
((exp(T_cosmo) - (T_cosmo ...
                                + T_cosmo)) / 0.59068))
14      5.223e-02  1.157e-01  y = inv((exp(exp(T_cosmo) - 5.082) + (1.8758
- T_cosmo)) + ...
                                (Anthropic * 0.11245))
16      5.204e-02  1.768e-03  y = inv(exp(exp(T_cosmo) - 5.082) +
(((0.11245 * Anthropic...
                                ) + 1.0052) * (1.8758 - T_cosmo)))
18      5.160e-02  4.321e-03  y = inv(((Anthropic * 0.11245) +
exp(exp(T_cosmo) - 5.082)...
                                ) + ((1.8758 - (T_cosmo +
normalized_density)) * 1.0052))
29      4.995e-02  2.954e-03  y = inv((((exp(T_cosmo - 161.1) + 1.0492) +
(Anthropic * ...
                                0.059005)) + exp(local_betti_2)) *
(exp(exp(T_cosmo) - 5.0...
                                909) + (1.9243 - T_cosmo))) +
(exp(local_betti_2) - normal...
                                ized_density))

```

Press 'q' and then <enter> to stop execution early.

Expressions evaluated per second: 9.840e+03
Progress: 1150 / 1240 total iterations (92.742%)

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.701e-02	9.236e-02	y = inv((Anthropic * 0.24972) + exp(T_cosmo))
8	6.927e-02	1.058e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.280e-02	6.350e-02	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo) - normaliz... ed_density)
12	5.958e-02	2.637e-02	y = inv((Anthropic * 0.087197) + exp(exp(T_cosmo + -0.505)... - T_cosmo))

```

13          5.863e-02  1.600e-02  y = inv((Anthropic * 0.10932) +
((exp(T_cosmo) - (T_cosmo ...
                                + T_cosmo)) / 0.59068))
14          5.220e-02  1.162e-01  y = inv((Anthropic * 0.10752) +
((exp(exp(T_cosmo) - 5.115...
                                ) + 1.9172) - T_cosmo))
16          5.204e-02  1.484e-03  y = inv(exp(exp(T_cosmo) - 5.082) +
(((0.11245 * Anthropic...
                                ) + 1.0052) * (1.8758 - T_cosmo)))
18          5.160e-02  4.321e-03  y = inv(((Anthropic * 0.11245) +
exp(exp(T_cosmo) - 5.082)...
                                ) + ((1.8758 - (T_cosmo +
normalized_density)) * 1.0052))
21          5.135e-02  1.568e-03  y = inv(((1.6678 - T_cosmo) +
(exp(local_betti_2) + exp(ex...
                                p(T_cosmo) - 5.1349))) * (((Anthropic *
0.063351) + 1.5813...
                                ) - normalized_density))
24          5.109e-02  1.700e-03  y = inv((1.0492 + ((0.059005 * Anthropic) +
exp(local_bett...
                                i_2))) * ((exp(exp(T_cosmo) - 5.0909) +
(1.9243 - T_cosmo)...
                                ) + exp(local_betti_2 -
normalized_density)))
25          5.017e-02  1.815e-02  y = exp(local_betti_2) +
inv(exp(exp(local_betti_2) - norm...
                                alized_density) + (exp(exp(T_cosmo) -
5.153) + ((1.5482 - ...
                                T_cosmo) * ((Anthropic * 0.065861) +
0.80938))))
27          5.013e-02  4.169e-04  y = inv(exp(exp(local_betti_2 - Anthropic) -
normalized_de...
                                nsity) + (exp(exp(T_cosmo) - 5.153) +
((1.5482 - T_cosmo) ...
                                * ((Anthropic * 0.065861) + 0.80938)))) +
exp(local_betti_...
                                2)
29          4.942e-02  7.153e-03  y = (exp(local_betti_2) +
inv(exp(exp(local_betti_2 - norm...
                                alized_density) - normalized_density) +
(exp(exp(T_cosmo) ...
                                - 5.153) + ((1.5482 - T_cosmo) *
((Anthropic * 0.065861) +...
                                0.80938)))))) / 0.83554

```

Press 'q' and then <enter> to stop execution early.

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	y = 0.13595
4	9.457e-02	2.550e-03	y = inv(T_cosmo + 6.6707)
5	8.907e-02	5.992e-02	y = inv(exp(T_cosmo) + Anthropic)
6	8.446e-02	5.319e-02	y = inv((T_cosmo * T_cosmo) + 3.5669)
7	7.701e-02	9.236e-02	y = inv((Anthropic * 0.24972) + exp(T_cosmo))
8	6.927e-02	1.058e-01	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo))
9	6.692e-02	3.455e-02	y = inv((Anthropic * 0.15882) + (exp(T_cosmo) - T_cosmo))
10	6.280e-02	6.350e-02	y = inv(exp((exp(T_cosmo) - T_cosmo) - T_cosmo) - normaliz... ed_density)
12	5.958e-02	2.637e-02	y = inv((Anthropic * 0.087197) + exp(exp(T_cosmo + -0.505)... - T_cosmo))
13	5.863e-02	1.600e-02	y = inv((Anthropic * 0.10932) + ((exp(T_cosmo) - (T_cosmo ... + T_cosmo)) / 0.59068))
14	5.220e-02	1.162e-01	y = inv((Anthropic * 0.10752) + (exp(exp(T_cosmo) - 5.115...)+ 1.9172) - T_cosmo))
16	5.204e-02	1.484e-03	y = inv(exp(exp(T_cosmo) - 5.082) + (((0.11245 * Anthropic...)+ 1.0052) * (1.8758 - T_cosmo)))
18	5.160e-02	4.321e-03	y = inv(((Anthropic * 0.11245) + exp(exp(T_cosmo) - 5.082)...)+ ((1.8758 - (T_cosmo + normalized_density)) * 1.0052))
20	4.941e-02	2.164e-02	y = inv((exp((T_cosmo + exp(T_cosmo)) - 6.607) + ((2.0932 ... - T_cosmo) * ((Anthropic * 0.060392) + 0.93443))) - normal... ized_density)
29	4.910e-02	6.906e-04	y = inv(exp(local_betti_2) + (exp(exp(local_betti_2 - norm... alized_density) - normalized_density) + (exp(exp(T_cosmo) ... - 5.153) + ((1.5482 - T_cosmo) * (Anthropic *0.065861) +... 0.80938)))))) / 0.83554

Press 'q' and then <enter> to stop execution early.

Complexity	Loss	Score	Equation
1	9.530e-02	0.000e+00	$y = 0.13595$
4	9.457e-02	2.550e-03	$y = \text{inv}(T_{\text{cosmo}} + 6.6707)$
5	8.907e-02	5.992e-02	$y = \text{inv}(\exp(T_{\text{cosmo}}) + \text{Anthropic})$
6	8.446e-02	5.319e-02	$y = \text{inv}((T_{\text{cosmo}} * T_{\text{cosmo}}) + 3.5669)$
7	7.701e-02	9.236e-02	$y = \text{inv}((\text{Anthropic} * 0.24972) + \exp(T_{\text{cosmo}}))$
8	6.927e-02	1.058e-01	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}))$
9	6.692e-02	3.455e-02	$y = \text{inv}((\text{Anthropic} * 0.15882) + (\exp(T_{\text{cosmo}})$ $- T_{\text{cosmo}}))$
10	6.280e-02	6.350e-02	$y = \text{inv}(\exp((\exp(T_{\text{cosmo}}) - T_{\text{cosmo}}) -$ $T_{\text{cosmo}}) - \text{normaliz...}$ $\text{ed_density})$
12	5.958e-02	2.637e-02	$y = \text{inv}((\text{Anthropic} * 0.087197) +$ $\exp(\exp(T_{\text{cosmo}} + -0.505)) \dots$ $- T_{\text{cosmo}}))$
13	5.863e-02	1.600e-02	$y = \text{inv}((\text{Anthropic} * 0.10932) +$ $((\exp(T_{\text{cosmo}}) - (T_{\text{cosmo}} \dots$ $+ T_{\text{cosmo}})) / 0.59068))$
14	5.220e-02	1.162e-01	$y = \text{inv}((\text{Anthropic} * 0.10752) +$ $((\exp(\exp(T_{\text{cosmo}}) - 5.115 \dots$ $) + 1.9172) - T_{\text{cosmo}}))$
16	5.204e-02	1.484e-03	$y = \text{inv}(\exp(\exp(T_{\text{cosmo}}) - 5.082) +$ $((0.11245 * \text{Anthropic} \dots$ $) + 1.0052) * (1.8758 - T_{\text{cosmo}}))$
18	5.160e-02	4.321e-03	$y = \text{inv}(((\text{Anthropic} * 0.11245) +$ $\exp(\exp(T_{\text{cosmo}}) - 5.082)) \dots$ $) + ((1.8758 - (T_{\text{cosmo}} +$ $\text{normalized_density})) * 1.0052))$
20	4.941e-02	2.164e-02	$y = \text{inv}((\exp((T_{\text{cosmo}} + \exp(T_{\text{cosmo}})) -$ $6.607) + ((2.0932 \dots$ $- T_{\text{cosmo}}) * ((\text{Anthropic} * 0.060392) +$ $0.93443))) - \text{normal...}$ $\text{ized_density})$
24	4.940e-02	7.677e-05	$y = \text{inv}(((0.69674 + (\text{Anthropic} * 0.088435))$ $* (1.486 - T_{\text{...}}$ $\text{cosmo})) + \exp(\exp(T_{\text{cosmo}}) - 5.164)) +$ $\exp(\exp(\text{local_betti} \dots$ $_2 - \text{normalized_density}) -$ $\text{normalized_density}))$
25	4.839e-02	2.061e-02	$y = \text{inv}(((0.69674 + (\text{Anthropic} * 0.088435)) *$ $(1.486 - T_{\text{c...}}$

```

osmo)) + ((inv(local_betti_2) +
exp(exp(T_cosmo) - 5.164))...
normalized_density)))
27      4.832e-02  6.600e-04  y = inv(exp(exp(T_cosmo) - 5.164) +
((((Anthropic * 0.088...
(inv(local_betti_2)...
normalized_density...
435) + 0.69674) * (1.486 - T_cosmo)) +
/ 0.81654)) + exp(exp(local_betti_2) -
)))
29      4.831e-02  1.233e-04  y = inv((((Anthropic * 0.088435) + 0.69674)
* (1.486 - T_...
cosmo)) + (inv(local_betti_2) / 0.81654))
+ (exp(exp(local...
_betti_2 - normalized_density) -
normalized_density) + exp...
(exp(T_cosmo) - 5.164)))
30      4.816e-02  3.185e-03  y = inv((exp(exp(T_cosmo) - 5.0878) +
exp((local_betti_2 +...
((exp(local_betti_2) -
normalized_density) * 5.1065)) + 1...
.0732)) + ((2.0825 - T_cosmo) *
(((Anthropic + T_cosmo) * ...
0.056117) + 0.89791)))

```

--- Top Candidate Equations ---

```

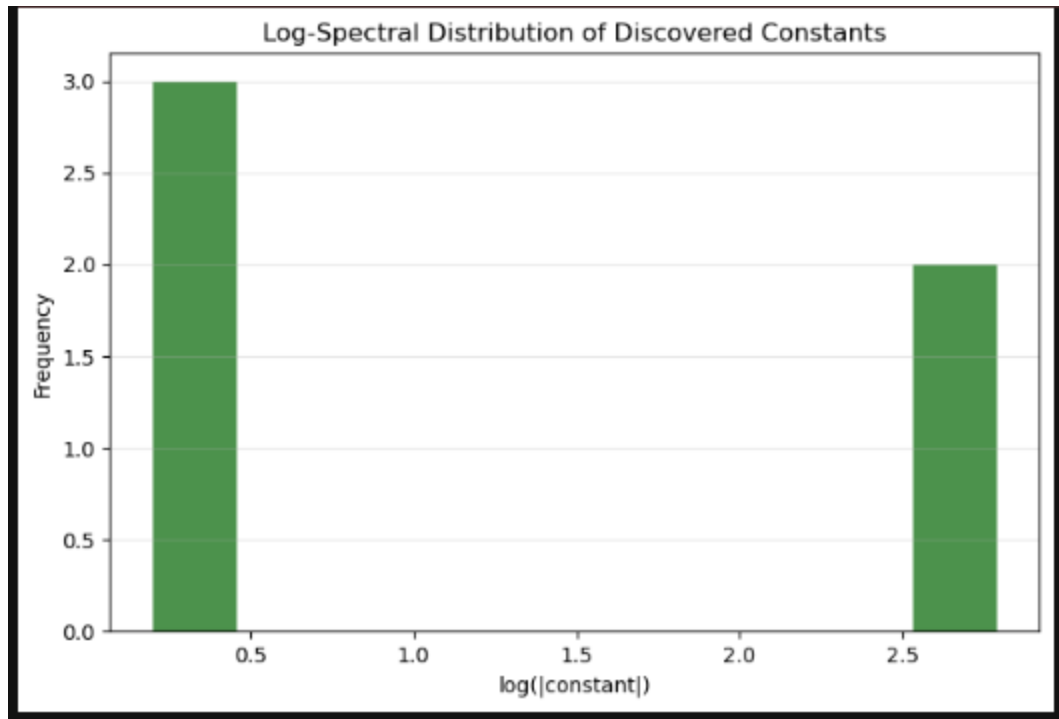
1/(0.1075172400000000*Anthropic - T_cosmo + e^(e^T_cosmo - 5.11496200000000) +
1.917238800000000)

```

Histogram of log(discovered constants)...

[Info: Final population:

[Info: Results saved to:



S.T.A.R. Pipeline Complete. Visualization and Foliation test saved.
- outputs/20260413_232621_Q7MTuI/hall_of_fame.csv